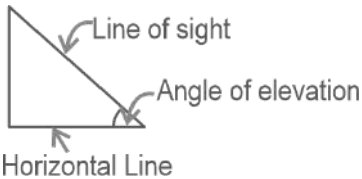


Height and Distance

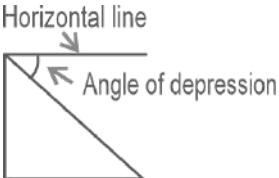
In this chapter following concepts / terminologies will be used.

Angle Of Elevation



The angle made by the line of sight with the horizontal line is called the angle of elevation.

Angle Of Depression



The angle made by the line of sight with the horizontal line is called the angle of depression.

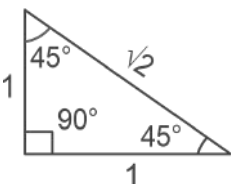
Note: When we come near object, the angle of elevation increases and when we go away from the object, the angle of elevation decreases.

Note: Angle of elevation and angle of depression are always equal.

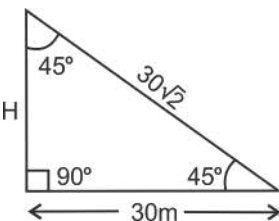
In this chapter trigonometric ratio and concepts will be used which we have discussed in detail in the trigonometry chapter.

Some important relation

Case :- (1)



Example :- The angle of elevation of the top of a tower from a point on the ground, 45° which is 30 m from the foot of the tower. Find the height of the tower?



Let the height of the tower be 'H'

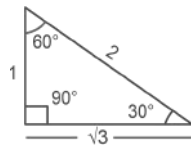
$$\tan 45^\circ = \frac{H}{30}$$

$$1 = \frac{H}{30} \Rightarrow H = 30 \text{ m}$$

Short Trick :- $1 \rightarrow 30 \text{ m}$

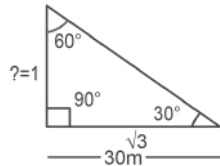
$$1 \rightarrow ? \Rightarrow ? = 30 \text{ m}$$

Case :- (2)



Example :- The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower is 30° . Find the height of the tower?

Solution :-



Let, the height of the tower be 'H'

$$\tan 30^\circ = \frac{H}{30}$$

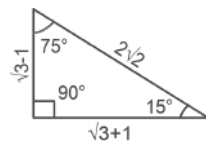
$$\frac{1}{\sqrt{3}} = \frac{H}{30} \Rightarrow H = \frac{30}{\sqrt{3}} \text{ m} = 10\sqrt{3} \text{ m}$$

Short Trick :-

$$\sqrt{3} \rightarrow 30 \text{ m}$$

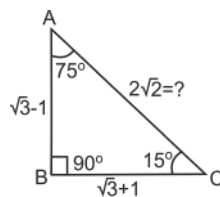
$$1 \rightarrow ? \Rightarrow ? = \frac{30}{\sqrt{3}} \text{ m} = 10\sqrt{3} \text{ m}$$

Case :- (3)



Example :- The angle of elevation of a ladder leaning against a wall is 15° and the foot of the ladder is 4 m away from the wall. The length of the ladder is :

Solution :-



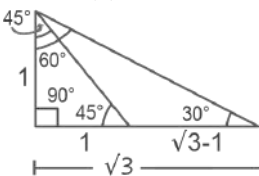
Let AB be the wall and AC be the ladder.

Then, $\angle ACB = 15^\circ$ and $BC = 4 \text{ m}$.

$$\sqrt{3} + 1 \rightarrow 4 \text{ m}$$

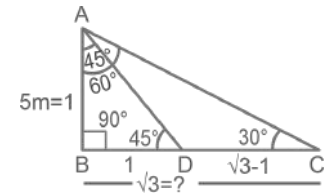
$$2\sqrt{2} \rightarrow ? \Rightarrow ? = \frac{8\sqrt{2}}{\sqrt{3} + 1}$$

Case :- (4)



Example :- A man on the top of a vertical observation tower observes a car moving at a uniform speed coming directly towards it and the angle of depression to change from 30° to 45° . If the height of the tower is 5 m then find the value of the total covered distance by the car?

Solution :-



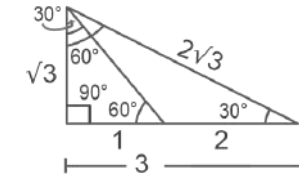
Let $AB = \text{Height of tower} = 1$ And total covered distance by the car $= BC = \sqrt{3}$

So, ATQ

$$1 = 5 \text{ m}$$

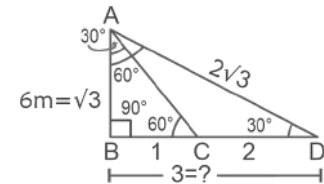
$$\sqrt{3} = ? \Rightarrow 5\sqrt{3}$$

Case :- (5)



Example :- A man on the top of a vertical observation tower observes a car moving at a uniform speed coming directly towards it and the angle of depression to change from 30° to 60° . If the height of the tower is 6 m then find the value of the total covered distance by the car?

Solution :-



Let $AB = \text{Height of tower} = \sqrt{3}$

And total covered distance by the car $= BD = 3$

So, A/Q

$$\sqrt{3} = 6 \text{ m}$$

$$3 = ? \Rightarrow 6\sqrt{3} \text{ m}$$

Variety Questions

Q.1. From a point A on the ground, the angle of elevation to the top of a building is 30° . From a point B, which is directly between A and the foot of the building, the angle of elevation is 45° . What is the ratio of the distance of point A from the building's foot to the distance of point B from the building's foot?

SSC CGL 19/09/2025 (Shift 2)

(a) $\sqrt{2} : 3$ (b) $\sqrt{3} : 1$ (c) $\sqrt{3} : \sqrt{2}$ (d) $2 : 3$

Q.2. A person was standing on a road near a mall. He was 1220 m away from the mall and able to see the top of the mall from the road in such a way that the top of a tree, which is in between him and the mall, was exactly in line of sight with the top of the mall. The tree height



is 40m and it is 80m away from him. How tall (in m) is the mall?

Graduate Level 31/07/2025 (Shift - 3)

(a) 605 (b) 610 (c) 600 (d) 675

Q.3. From a point on a bridge built over a river, the angles of depression to the two opposite banks are 30° and 45° respectively. If the width of the river is $80(\sqrt{3} + 1)$ metres, find the height of the bridge above the river banks.

Graduate Level 26/07/2025 (Shift - 3)

(a) 80m (b) 60m (c) 40 m (d) 65 m

Q.4. An observer on a tower 45 meters high spots a car. The angle of depression to the car is 45° . What is the distance of the car from the base of the tower?

Higher Secondary 25/07/2025 (Shift - 2)

(a) $45\sqrt{3}$ m (b) $45\sqrt{2}$ m
(c) 45 m (d) 90 m

Q.5. A man looks at the reflection of the top of the lamp-post on the mirror that is 6.6 m away from the foot of the lamp-post. The man's height is 1.25 m and he is standing 2 m away from the mirror. Assuming that the mirror is placed on the ground, facing the sky and the man, and that the mirror and the lamp-post are in a same line, find the height of the lamp-post (in metres).

(a) 3.97 (b) 4.28 (c) 4.45 (d) 4.13

Q.6. The shadow of a tower standing on level ground is found to be 40 m longer when the Sun's altitude is 30° than when it was 45° . Find the height (in meters) of the tower.

(a) $10(\sqrt{3} - 1)$ (b) $20(\sqrt{3} - 1)$
(c) $20(\sqrt{3} + 1)$ (d) $10(\sqrt{3} + 1)$

Q.7. Two pillars of equal height stand on either side of a roadway which is 150 m wide. At a point in the road between pillars, the elevations of the pillars are x° and y° so that $\tan x^\circ = \frac{2}{5}$, $\tan y^\circ = \frac{3}{5}$, then the height of each pillar is :

(a) 39 m (b) 36 m (c) 42 m (d) 33 m

Q.8. Subhas, a 3.15 m tall tree, and an 11.25 m tall building are positioned such that their feet on the ground are collinear and the tree is located between Subhas and the building. The tree is located at a distance of 7.5 m from Subhas and at a distance of 45 m from the building. Further, the eyes of Subhas, the top of the tree, and the top of the building fall in one line. Find the height (in m) from the ground at which Subhas' eyes are situated.

(a) 1.75 (b) 1.8 (c) 1.6 (d) 1.5

Q.9. From the top of an upright pole 17.75 m high, the angle of elevation of the top of an upright tower was 60° . If the tower was 57.75 m tall, how far away (in m) from the foot of the pole was the foot of the tower?

(a) $40\sqrt{3}$ (b) $\frac{151\sqrt{3}}{6}$ (c) $\frac{77}{4}\sqrt{3}$ (d) $\frac{40\sqrt{3}}{3}$

Q.10. A pole 23 m long reaches a window which is $3\sqrt{5}$ m above the ground on one side of a street. Keeping its foot at the same point, the pole is turned to the other side of the street to reach a window $4\sqrt{15}$ m high. What is the width (in m) of the street?

(a) 17 (b) 35 (c) 39 (d) 22

Q.11. A person was standing on a road near a mall. He was 1215 m away from the mall and able to see the top of the mall from the road in such a way that the top of a tree, which is in between him and the mall, was exactly in line of sight with the top of the mall. The tree height is 20m and it is 60m away from him. How tall (in m) is the mall?

(a) 405 (b) 250 (c) 300 (d) 375

Examination wise Questions

SSC CGL 2025 Tier - I

Q.12. A vertical pole of height H stands on the ground. From point P, the angle of elevation of the top is 45° . From point Q, 20 meters away from P, the angle of elevation is 30° . What is the height of the pole ?

SSC CGL 15/09/2025 (Shift 2)

(a) $5(\sqrt{3} + 1)$ m (b) $10(\sqrt{3} + 1)$ m
(c) 10 m (d) 20 m

Q.13. A tower stands 60 meters tall. From its peak, the angles of depression to the top and bottom of a nearby building are measured at 45° and 60° , respectively. Determine the approximate height of the building as well as the horizontal distance separating the tower from the building.

SSC CGL 16/09/2025 (Shift 2)

(a) height = 21 m, distance = 50 m
(b) height = 25 m, distance = 35 m
(c) height = 28 m, distance = 15 m
(d) height = 18 m, distance = 15 m

Q.14. A flagpole has a shadow measuring 20 meters. Given that the height of the flagpole is $20\sqrt{3}$ meters, What is the angle at which the sun is elevated ?

SSC CGL 17/09/2025 (Shift 1)

(a) 45° (b) 30° (c) 60° (d) 90°

Q.15. A steel cable is tied from the top of a vertical flagpole to a point on the ground 20 meters away from its base. If the cable makes an angle of 60° with the ground, find the length of the cable?

SSC CGL 17/09/2025 (Shift 3)

(a) 10 m (b) $10\sqrt{3}$ m (c) 40 m (d) $20\sqrt{3}$ m

Q.16. A man standing on the top of a 150 m high tower observes the angles of depression of two cars on the same sides of the tower to be 45° and 60° . What is the distance between the two cars?

SSC CGL 18/09/2025 (Shift 1)

(a) $150(\sqrt{3} - 1)$ m (b) $150\sqrt{3}$ m
(c) $150(1 - \frac{1}{\sqrt{3}})$ m (d) $150(1 + \frac{1}{\sqrt{3}})$ m

Q.17. A vertical tree PQ has its base on the ground Q. From a point A 20 m away from Q, the angle of elevation of P is x° . From another point B 60 m away from Q (and in the same line as A), the angle of elevation of P is y° . If the height of the tree is $20\sqrt{3}$ m, what is the value of $x^\circ + y^\circ$?

SSC CGL 18/09/2025 (Shift 1)

(a) 60° (b) 75° (c) 90° (d) 120°

Q.18. A rock climber uses a rope to ascend a cliff. If the vertical height of the cliff is 23 meters and the rope makes an angle of 45° with the level ground, what is the minimum length of the rope required to reach the top ?

SSC CGL 18/09/2025 (Shift 2)

(a) 46 m (b) $23\sqrt{2}$ m (c) 60 m (d) $46\sqrt{2}$ m

Q.19. A tower generates a height that is 60 meters long when the sun's angle of elevation is 60 degrees. What is the shadow of the tower ?

SSC CGL 18/09/2025 (Shift 2)

(a) $45\sqrt{3}$ m (b) $38\sqrt{2}$ m
(c) $20\sqrt{3}$ m (d) $19\sqrt{2}$ m

Q.20. An airplane flying at a height of 400 m notices the angles of depression to two boats located on opposite sides as 45° and 30° . Find the distance between the two boats.

SSC CGL 18/09/2025 (Shift 3)

(a) $400(\sqrt{3} - 1)$ m (b) $400\sqrt{3}$ m
(c) $400(1 + \frac{1}{\sqrt{3}})$ m (d) $400(\sqrt{3} + 1)$ m

Q.21. A statue that is 3.6 meters in height is positioned on top of a pedestal. From a location on the ground, the angle of elevation to the top of the statue measures 60 degrees, while the angle of



elevation to the top of the pedestal from the same point is 45 degrees. Determine the height of the pedestal.

SSC CGL 18/09/2025 (Shift 3)

- (a) $1.5(\sqrt{3} - 1)$ m (b) $1.8(\sqrt{3} - 1)$ m
(c) $1.6(\sqrt{3} + 1)$ m (d) $1.8(\sqrt{3} + 1)$ m

Q.22. A tree is 40 meters tall. What is the length of the shadow cast by the tower when the sun's elevation is 30° ?

SSC CGL 19/09/2025 (Shift 1)

- (a) 20 m (b) 30 m (c) $40\sqrt{3}$ m (d) $30\sqrt{3}$ m

Q.23. A wheelchair ramp needs to reach a platform that is 2.5 meters high. If the ramp is designed to make an angle of 30° with the level ground, what is the length of the ramp?

SSC CGL 19/09/2025 (Shift 2)

- (a) 1.5 m (b) 2 m (c) 5 m (d) 3 m

Q.24. A ladder reaches a window 9 meters high on a vertical wall. If the ladder is positioned at a 30° angle to the ground, what is its length?

SSC CGL 19/09/2025 (Shift 3)

- (a) 18 m (b) 20 m (c) 32 m (d) 12 m

Q.25. A 7-foot person casts a 5-foot shadow. At the same time, a tree casts a 35-foot shadow. How tall is the tree?

SSC CGL 20/09/2025 (Shift 2)

- (a) 40 feet (b) 49 feet
(c) 50 feet (d) 30 feet

Q.26. A kite is soaring at an altitude of 45 meters above the ground, with its string temporarily secured to a point on the ground. The angle of elevation of the string relative to the ground is 30 degrees. Given that the string remains taut, what is the length of the string?

SSC CGL 23/09/2025 (Shift 2)

- (a) 60 m (b) $45\sqrt{3}$ m (c) 90 m (d) $55\sqrt{3}$ m

Q.27. A man observes the top of a tower at an angle of elevation of 30° . He walks 20 meters towards the tower, and the angle of elevation becomes 45° . What is the height of the tower?

SSC CGL 24/09/2025 (Shift 1)

- (a) $10(\sqrt{3} + 1)$ m (b) $10(\sqrt{3} + 3)$ m
(c) $15(\sqrt{3} + 1)$ m (d) $15(3 + \sqrt{3})$ m

Q.28. Two ships positioned on opposite sides of a lighthouse observe the angles of elevation to the top of the lighthouse, which are 30° and 45° respectively. Given that the height of the lighthouse is 200 meters, determine the distance between the two ships.

SSC CGL 24/09/2025 (Shift 3)

- (a) $200(\sqrt{3} - 1)$ m (b) $\frac{200}{\sqrt{3}}$ m

- (c) $200(\frac{1}{\sqrt{3}} + 1)$ m (d) $200(\sqrt{3} + 1)$ m

Q.29. The angles of elevation of the top of a tower from two points on the ground, 400 m apart, and on opposite sides of the tower, are 30° and 45° . What is the height of the tower?

SSC CGL 26/09/2025 (Shift 2)

- (a) $100(\sqrt{3} + 1)$ m (b) $200(\sqrt{3} - 1)$ m
(c) $100(\sqrt{3} - 1)$ m (d) 200 m

Q.30. From a point on a bridge built over a river, the angles of depression to the two opposite banks are 30° and 45° respectively. If the width of the river is $50(\sqrt{3} + 1)$ metres, find the height of the bridge above the riverbanks.

SSC CGL 14/10/2025 (Shift 2)

- (a) 50 m (b) 45 m (c) 30 m (d) 55 m

Q.31. From the top of a 60 m tower, the angle of depression of two objects on the same side of the tower are 30° and 45° . Find the distance between the objects.

SSC CGL 14/10/2025 (Shift 2)

- (a) 21.13 m (b) 28.7 m
(c) 43.92 m (d) 58.87 m

Q.32. A flagpole has a shadow measuring 20 meters. Given that the height of the flagpole is $20\sqrt{3}$ meters, What is the angle at which the sun is elevated?

SSC CGL 14/10/2025 (Shift 3)

- (a) 30° (b) 45° (c) 60° (d) 75°

SSC Selection Post (XIII)

Q.33. Two observers stand 40 meters apart and observe the top of a tower. The angles of elevation are 30° and 45° . Find the height of the tower (assuming both observers are on the same side of the tower).

Higher Secondary 24/07/2025 (Shift - 2)

- (a) $20(\sqrt{3} - 1)$ m (b) $20(\sqrt{3} + 1)$ m
(c) $40(\sqrt{3} + 1)$ m (d) $20\sqrt{3}$ m

Q.34. From a point on the ground, the angle of elevation of the top of a tower is 30° . If the distance from the point to the base of the tower is 70 meters, find the height of the tower.

Higher Secondary 26/07/2025 (Shift - 2)

- (a) $70\sqrt{3}$ m (b) $\frac{(70\sqrt{3})}{3}$ m
(c) 70m (d) $\frac{(35\sqrt{3})}{2}$ m

Q.35. The angle of elevation of the top of a tower is 30° from a point on the ground. On moving 200 m closer to the

base, the angle becomes 60° . Find the height of the tower.

Higher Secondary 26/07/2025 (Shift - 2)

- (a) 50 m (b) 100 m
(c) $50\sqrt{3}$ m (d) $100\sqrt{3}$ m

Q.36. The angle of elevation of the top of a tower from a point 120 m away from its base is 45° . Find the height of the tower.

Higher Secondary 28/07/2025 (Shift - 2)

- (a) $120\sqrt{2}$ m (b) 120 m
(c) $60\sqrt{3}$ m (d) $\frac{60}{\sqrt{3}}$ m

Q.37. Two vertical poles have heights of 12 meters and 18 meters vertically on a plane ground. If the distance between their feet is 8m, the distance between their tops is:

Higher Secondary 28/07/2025 (Shift - 2)

- (a) 12 m (b) 10 m (c) 14 m (d) 9 m

Q.38. Two points P and Q are at distances x and y from the base of a tower. The angles of elevation of the top of the tower from P and Q are complementary. The height of the tower is:

Higher Secondary 29/07/2025 (Shift - 2)

- (a) $\sqrt{xy}$ (b) $\frac{\sqrt{x}}{y}$ (c) $\frac{\sqrt{y}}{x}$ (d) $\frac{x}{y}$

Q.39. The angle of elevation of the top of a tower from a point on the ground is 60° . If the tower is $40\sqrt{3}$ meters high, find the distance of the point from the base of the tower.

Higher Secondary 30/07/2025 (Shift - 2)

- (a) 40 m (b) $40\sqrt{3}$ m
(c) $50\sqrt{3}$ m (d) $20\sqrt{3}$ m

Q.40. A balloon reaches a vertical height of 90 m from the ground. The string attached to the balloon is inclined at 45° to the horizontal. The length of the string is:

Higher Secondary 30/07/2025 (Shift - 2)

- (a) $60\sqrt{2}$ m (b) $140\sqrt{2}$ m
(c) $\frac{90}{\sqrt{2}}$ m (d) $90\sqrt{2}$ m

Q.41. A vertical tower casts its gaze upwards from the ground. If you stand 15 meters from its base and look up at a 30-degree angle to its peak, how tall is the tower?

Higher Secondary 31/07/2025 (Shift - 2)

- (a) 30 m (b) $10\sqrt{3}$ m
(c) $15\sqrt{3}$ m (d) $5\sqrt{3}$ m

Q.42. A student is standing, reading an article 28 meters away from his school building. If the angle of elevation of the top of the school building is 45 degrees, how tall is the school building?

Higher Secondary 02/08/2025 (Shift - 2)



(a) 28 m (b) 33 m (c) 46 m (d) 39 m

Q.43. An airplane in level flight at an altitude of 3 km from the ground is observed at an angle of elevation of 60° . After 20 seconds, its angle of elevation is observed to be 30° . The speed of the airplane in km/hr is:

Higher Secondary 29/08/2025 (Shift - 2)

(a) $\frac{540}{\sqrt{3}}$ (b) $190\sqrt{3}$ (c) $\frac{360}{\sqrt{3}}$ (d) $360\sqrt{3}$

Q.44. An aircraft is flying at a steady altitude of $1500\sqrt{3}$ meters. From a point on the ground, the angle of elevation of the airplane changes from 60° to 30° in 15 seconds as it flies directly away from the observer. What is the approximate speed of the airplane?

Graduate Level 24/07/2025 (Shift - 3)

(a) 150 m/s (b) 200 m/s
(c) 250 m/s (d) 300 m/s

Q.45. A hot-air balloon directly above point P is rising vertically at 4 m/s. An observer at point Q, 400 m due east of P, watches the balloon. At the moment when the angle of elevation is 45° , how long until it reaches an angle of elevation of 60° ? (Assume no horizontal drift.)

Graduate Level 25/07/2025 (Shift - 3)

(a) 73.2 s (b) 25.8 s
(c) 55.2 s (d) 78.5 s

Q.46. From the summit of a 200 m high cliff, the angles of depression to two boats situated on the same side of the cliff are measured at 45° and 60° respectively. The boats are moving towards the cliff in a straight line. After 10 minutes, the angle of depression to the first boat becomes 60° , while the angle to the second boat becomes 75° . If both boats move at constant but different speeds, determine the initial distance between the two boats.

Graduate Level 25/07/2025 (Shift - 3)

(a) $200\left(\frac{\sqrt{3}-1}{\sqrt{3}}\right)$ m (b) $200\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)$ m
(c) $100\left(\frac{\sqrt{3}-1}{\sqrt{3}}\right)$ m (d) $100\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)$ m

Q.47. From the top of a 180 m high cliff, an observer notices a car moving away from the base of the cliff. The angle of depression of the car changes from 60° to 45° in 15 seconds. What is the approximate speed of the car?

Graduate Level 29/07/2025 (Shift - 3)

(a) 6.03 m/s (b) 7.08 m/s
(c) 5.07 m/s (d) 8.05 m/s

Q.48. A man on a lighthouse that stands $60\sqrt{3}$ meters tall is observing a boat moving away from it. Over the course of

15 seconds, the angle of depression to the boat shifts from 60° to 30° . What is speed of the boat?

Graduate Level 02/08/2025 (Shift - 3)

(a) 5 m/s (b) 8 m/s
(c) 6 m/s (d) 7 m/s

Q.49. A ladder leaning against a wall makes an angle α with the horizontal ground such that $\tan \alpha = \frac{3}{4}$. If the foot of the ladder is 5 m away from the wall, what is the length (in m) of the ladder?

Graduate Level 29/08/2025 (Shift - 1)

(a) 5.25 (b) 3.75 (c) 6.25 (d) 4.5

eduquity-based pattern (ebp) Questions

Q.50. A pole 10 m long rests slantly against a vertical wall AB making an angle 30° with the horizontal (ground). Find how far the foot of the pole is from the wall (in metres).

(a) 15 (b) $\frac{5}{\sqrt{3}}$ (c) 10 (d) $5\sqrt{3}$

Q.51. A mirror is placed on the ground facing upwards. A man sees the top of a tower in the mirror which is at a distance of 105 m from the mirror. The man is 0.5 m away from the mirror, and his height is 1.5 m. Find the height of the tower (in metres).

(a) 280 (b) 115 (c) 315 (d) 210

Q.52. A vertical pole of length 6 m casts a 4 m long shadow on the ground. At the same time, a tower casts a 28 m long shadow. Find the height of the tower.

(a) 48 m (b) 32 m (c) 36 m (d) 42 m

Q.53. A pole stands vertically on the ground with the help of a 12-metre steel wire tied to its top and affixed on the ground. If the steel wire makes an angle of 30° with the horizontal ground, then the height of the pole is equal to:

(a) 8 meters (b) $6\sqrt{3}$ meters
(c) $8\sqrt{3}$ meters (d) 6 meters

Q.54. A man observed the top of an electric pole at angles of elevation of 60° and 30° , respectively, from two different observation points P and Q. If the distance between P and Q is 10 m, find the height of the pole (in m).

(a) $5\sqrt{3}$ (b) 5 (c) 15 (d) $15\sqrt{3}$

Q.55. From the top of a tower, the angle of depression of the top of a 10 m high building is 60° . If the distance between the tower and the building is $50\sqrt{3}$ m, find the height of the tower.

(a) 150 m (b) 100 m (c) 160 m (d) 140 m

Q.56. A vertical pillar 42 cm long casts a 35 cm long shadow. At the same time, a tower casts a shadow 25 m long. Find the height of the tower.

(a) 28 m (b) 35 m (c) 32 m (d) 30 m

Q.57. Suhas, a 3.15 m tall tree, and a building are positioned such that their feet on the ground are collinear and the tree is located between Suhas and the building. The tree is located at a distance of 7.5 m from Suhas and at a distance of 45 m from the building. Further, the eyes of Suhas, the top of the tree, and the top of the building fall in one line, and the eyes of Suhas are at a height of 1.8 m from the ground. Find the height (in m) of the building.

(a) 11.25 (b) 11.50 (c) 10.25 (d) 10.75

Q.58. Suhas, a tree, and an 11.25 m tall building are positioned such that their feet on the ground are collinear and the tree is located between Suhas and the building. The tree is located at a distance of 7.5 m from Suhas and at a distance of 45 m from the building. Further, the eyes of Suhas, the top of the tree, and the top of the building fall in one line, and the eyes of Suhas are at a height of 1.8 m from the ground. Find the height (in m) of the tree.

(a) 3.00 (b) 3.30 (c) 3.15 (d) 3.45

Q.59. A person standing at a distance looks at a building having a height of 1000 metres. The angle between the top of the building and the ground is 30° . At what approximate distance (in metres) is the person standing away from the building.

(a) 1000 (b) 936 (c) 1732 (d) 1542

Q.60. A vertical pole of 28m height casts a 19.2m long shadow. At the same time, find the length of the shadow cast by another pole of 52.5m height.

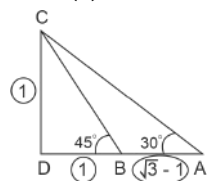
(a) 36 m (b) 35 m (c) 40 m (d) 30 m

Answer Key :-

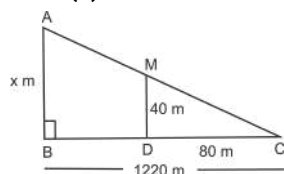
1.(b)	2.(b)	3.(a)	4.(c)
5.(d)	6.(c)	7.(b)	8.(b)
9.(d)	10.(c)	11.(a)	12.(b)
13.(b)	14.(c)	15.(c)	16.(c)
17.(c)	18.(b)	19.(c)	20.(d)
21.(d)	22.(c)	23.(c)	24.(a)
25.(b)	26.(c)	27.(a)	28.(d)
29.(b)	30.(a)	31.(c)	32.(c)
33.(b)	34.(b)	35.(d)	36.(b)



37.(b)	38.(a)	39.(a)	40.(d)
41.(d)	42.(a)	43.(d)	44.(b)
45.(a)	46.(a)	47.(c)	48.(b)
49.(c)	50.(d)	51.(c)	52.(d)
53.(d)	54.(a)	55.(c)	56.(d)
57.(a)	58.(c)	59.(c)	60.(a)

Solutions :-**Sol.1.(b)**

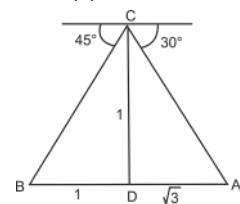
$$DA : DB \Rightarrow \sqrt{3} : 1$$

Sol.2.(b)

we have to find out the length of AB

$\triangle ABC \sim \triangle MDC$ (by AAA rule)

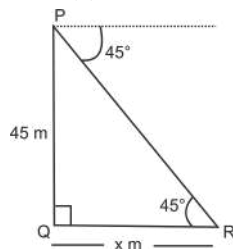
$$\frac{X}{40} = \frac{1220}{80} \Rightarrow x = \frac{1220 \times 40}{80} = 610 \text{ m}$$

Sol.3.(a)

Width of river = $BD + DA = BA$

$$\sqrt{3} + 1 = 80(\sqrt{3} + 1) \Rightarrow 1 \text{ unit} = 80 \text{ m}$$

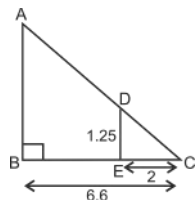
So, the height of bridge = 1 unit = 80 m

Sol.4.(c)

In $\triangle PQR$,

$$\tan 45^\circ = \frac{45}{x}$$

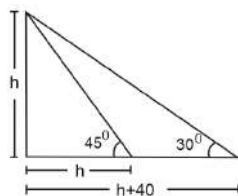
$$1 = \frac{45}{x} \Rightarrow x = 45 \text{ m}$$

Sol.5.(d)

$$\text{From similarity concept :- } \frac{EC}{ED} = \frac{BC}{AB}$$

$$\frac{2}{1.25} = \frac{6.6}{AB}$$

$$(\text{height of the lamp post}) AB = 3.3 \times 1.25 = 4.125 \text{ m (4.13 approx)}$$

Sol.6.(c)

$$\tan 30^\circ = \frac{h}{h+40}$$

$$\frac{1}{\sqrt{3}} = \frac{h}{h+40}$$

$$h+40 = \sqrt{3}h$$

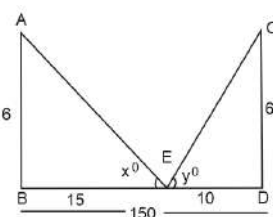
$$\therefore \text{height of tower (h)} = \frac{40}{\sqrt{3}-1}$$

$$= 20(\sqrt{3} + 1) \text{ meter}$$

Sol.7.(b) Let AB and CD be the two pillars of equal height

$$\tan x^\circ = \frac{AB}{BE} = \frac{2}{5} \text{ or } \frac{6}{15}$$

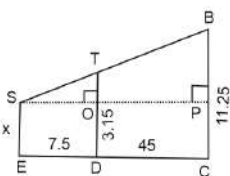
$$\tan y^\circ = \frac{CD}{DE} = \frac{3}{5} \text{ or } \frac{6}{10}$$



ATQ,

$$15 + 10 = 25 \text{ unit} \text{ ----- } 150 \text{ m}$$

$$\text{Then, 6 unit} \text{ ----- } \frac{150}{25} \times 6 = 36 \text{ m}$$

Sol.8.(b)

$$EC = SP = 7.5 + 45 = 52.5 \text{ metre}$$

$$BP = 11.25 - x \text{ metre}$$

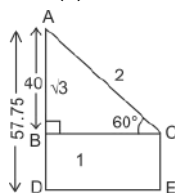
$$\triangle SOT \sim \triangle SPB$$

$$\frac{SO}{SP} = \frac{TO}{BP} \Rightarrow \frac{7.5}{52.5} = \frac{TO}{BP} \Rightarrow \frac{TO}{BP} = \frac{1}{7}$$

$$6 \text{ units} = (11.25 - 3.15) = 8.1 \text{ metre}$$

$$TO (1 \text{ unit}) = \frac{8.1}{6} = 1.35 \text{ metre}$$

$$SE = OD = (3.15 - 1.35) = 1.8 \text{ meter}$$

Sol.9.(d)

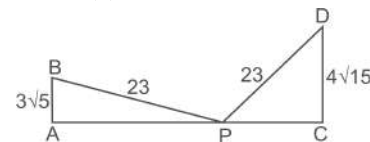
$$\text{Angle} \rightarrow 30^\circ : 60^\circ : 90^\circ$$

$$\text{Side} \rightarrow 1 : \sqrt{3} : 2$$

In $\triangle ABC$,

$$\sqrt{3} \text{ units} = 40 \text{ m} \Rightarrow 1 \text{ unit} = \frac{40\sqrt{3}}{3} \text{ m}$$

So, The foot of the tower is $\frac{40\sqrt{3}}{3} \text{ m}$ away from the foot of the pole.

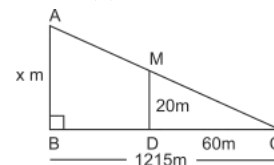
Sol.10.(c)

Let P be the fixed point of the pole and B and D are two windows.,

$$\text{In } \triangle ABP, AP = \sqrt{(23)^2 - (3\sqrt{5})^2} = \sqrt{529 - 45} = \sqrt{484} = 22 \text{ m.}$$

$$\text{In } \triangle DPC, PC = \sqrt{(23)^2 - (4\sqrt{15})^2} = \sqrt{529 - 240} = \sqrt{289} = 17 \text{ m.}$$

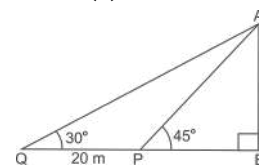
So, width of street = $22 + 17 = 39 \text{ m}$

Sol.11.(a)

we have to find out the length of AB

$\triangle ABC \sim \triangle MDC$ (by AAA rule)

$$\frac{X}{20} = \frac{1215}{60} \Rightarrow x = \frac{1215 \times 20}{60} = 405 \text{ m}$$

Sol.12.(b)

In $\triangle ABP$,

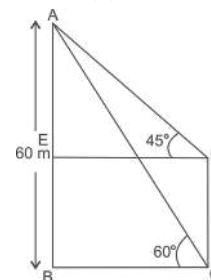
$$\Rightarrow \tan 45^\circ = \frac{H}{PB} \Rightarrow PB = H$$

In $\triangle ABQ$,

$$\Rightarrow \tan 30^\circ = \frac{H}{20 + PB} \Rightarrow \frac{1}{\sqrt{3}} = \frac{H}{20 + H}$$

$$\Rightarrow 20 + H = \sqrt{3}H \Rightarrow H = \frac{20}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1}$$

$$\Rightarrow H = 10(\sqrt{3} + 1) \text{ m}$$

Sol.13.(b)

In $\triangle ABC$,

$$\tan 60^\circ = \frac{60}{BC} \Rightarrow BC = \frac{60}{\sqrt{3}} = 34.68 \text{ mts}$$



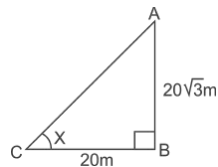
In $\triangle AED$,

$$\tan 45^\circ = \frac{60 - EB}{34.68}$$

$$1 = \frac{60 - EB}{34.68} \Rightarrow 34.68 = 60 - EB$$

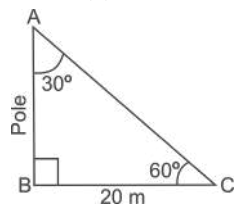
EB = DC [given]

$\therefore$ Height of building (DC) = 25.36 m, $\approx$ 25m, Distance = 34.68 m $\approx$ 35m

Sol.14.(c)In $\triangle ABC$,

$$\Rightarrow \tan X = \frac{AB}{BC} = \frac{20\sqrt{3}}{20}$$

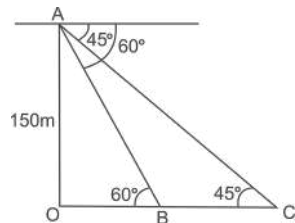
$$\Rightarrow \tan X = \sqrt{3} \Rightarrow X = 60^\circ$$

Sol.15.(c)Angle $\rightarrow 30^\circ : 60^\circ : 90^\circ$ Side $\rightarrow 1 : \sqrt{3} : 2$

According to the question,

1 unit = 20 meters $\Rightarrow$ 2 units = 40 meters

Length of the cable = 40 meters

Sol.16.(c)

Let, AO = tower length = 150 m

In $\triangle AOC$,

$$\Rightarrow \tan 45^\circ = \frac{150}{OC}$$

$$\Rightarrow OC = 150 \text{ m}$$

Now, in $\triangle AOB$,

$$\Rightarrow \tan 60^\circ = \frac{150}{OB}$$

$$\Rightarrow \sqrt{3} = \frac{150}{OB} \Rightarrow OB = \frac{150}{\sqrt{3}} \text{ m}$$

Hence, the distance between the two

$$\text{cars} = 150 - \frac{150}{\sqrt{3}} = 150 \left(1 - \frac{1}{\sqrt{3}} \right) \text{ m}$$

Sol.17.(c) According to the question,

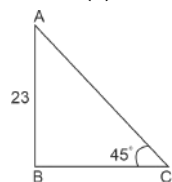
$$\tan x^\circ = \frac{\text{height}}{AQ} = \frac{20\sqrt{3}}{20} = \sqrt{3}$$

$$\Rightarrow x^\circ = 60^\circ$$

Now

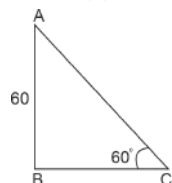
$$\tan y^\circ = \frac{\text{height}}{BQ} = \frac{20\sqrt{3}}{60} = \frac{\sqrt{3}}{3}$$

$$\Rightarrow y^\circ = 30^\circ$$

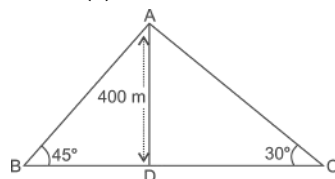
Hence, $x^\circ + y^\circ = 60^\circ + 30^\circ = 90^\circ$ **Sol.18.(b)**

$$\sin 45^\circ = \frac{AB}{AC}$$

$$\frac{1}{\sqrt{2}} = \frac{23}{AC} \Rightarrow AC = 23\sqrt{2}$$

Sol.19.(c)

$$\tan 60^\circ = \frac{60}{BC} \Rightarrow \sqrt{3} = \frac{60}{BC} \Rightarrow BC = 20\sqrt{3}$$

Sol.20.(d)

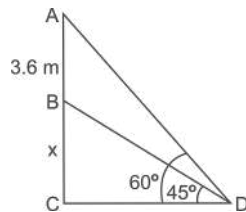
From the right triangles:

$$\tan 45^\circ = \frac{AD}{BD} \Rightarrow BD = 400 \text{ m}$$

$$\tan 30^\circ = \frac{AD}{CD} \Rightarrow CD = 400\sqrt{3} \text{ m}$$

Distance between boats:

$$BD + CD = 400 + 400\sqrt{3} = 400(1 + \sqrt{3}) \text{ m}$$

Sol.21.(d)

$$\tan \theta = \frac{\text{Perpendicular}}{\text{Base}}$$

$$\text{Ratio} \rightarrow \tan 60^\circ : \tan 45^\circ$$

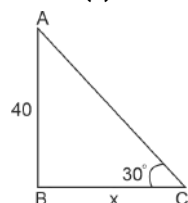
$$\frac{\sqrt{3}}{1} : \frac{1}{1}$$

According to question,

$$\Rightarrow (\sqrt{3} - 1) \text{ units} = 3.6$$

$$\Rightarrow 1 \text{ unit} = \frac{3.6}{\sqrt{3} - 1} \times 1 \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= 1.8(\sqrt{3} + 1) \text{ m}$$

So, the height of pedestal = $1.8(\sqrt{3} + 1) \text{ m}$ **Sol.22.(c)**

Let the shadow of the tree be x.

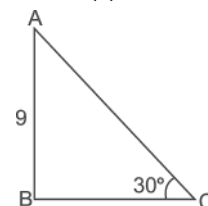
In triangle ABC

$$\tan 30^\circ = \frac{AB}{BC}$$

$$\frac{1}{\sqrt{3}} = \frac{40}{x} \Rightarrow x = 40\sqrt{3} \text{ m}$$

$$\text{Sol.23.(c)} \sin 30^\circ = \frac{2.5}{\text{Ramp length}}$$

$$\text{Ramp length} = \frac{2.5}{\sin 30^\circ} \Rightarrow \frac{2.5}{\frac{1}{2}} = 5 \text{ m}$$

Sol.24.(a)

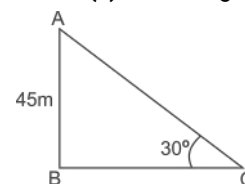
$$\sin 30^\circ = \frac{AB}{AC}$$

$$\frac{1}{2} = \frac{9}{AC} \Rightarrow AC = 18 \text{ m}$$

Sol.25.(b) 5ft shadow $\rightarrow$ 7foot

$$35 \text{ ft shadow} \rightarrow \frac{7}{5} \times 35 = 49 \text{ ft}$$

Height of tree = 49ft.

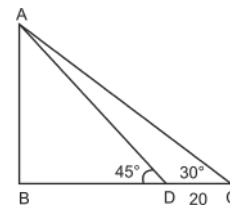
Sol.26.(c) According to diagram

In triangle ABC

$$\sin \theta = \frac{AB}{AC}$$

$$\sin 30^\circ = \frac{45}{AC}$$

$$\frac{1}{2} = \frac{45}{AC} \Rightarrow AC = 90 \text{ m}$$

Sol.27.(a)

Let BC = x metre

$$\tan 30^\circ = \frac{AB}{BC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{AB}{x}$$

$$x = \sqrt{3}AB \text{ ---- (i)}$$

$$\tan 45^\circ = \frac{AB}{BC - CD}$$

$$1 = \frac{AB}{x - 20} \Rightarrow x - 20 = AB$$

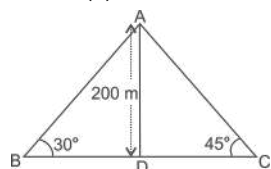
$$\sqrt{3}AB - 20 = AB$$

$$AB(\sqrt{3} - 1) = 20$$

$$AB = \frac{20}{\sqrt{3} - 1} \Rightarrow \frac{20}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= \frac{20(\sqrt{3} + 1)}{2} = 10(\sqrt{3} + 1) \text{ m}$$

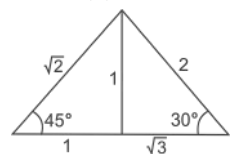


Sol.28.(d)

$$\tan 30^\circ = \frac{200}{BD} \Rightarrow BD = 200\sqrt{3},$$

$$\tan 45^\circ = \frac{200}{CD} \Rightarrow CD = 200$$

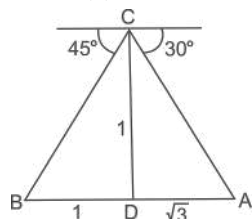
$$\text{Distance} = BD + DC = 200(\sqrt{3} + 1) \text{ m}$$

Sol.29.(b)

$$A/Q, (\sqrt{3} + 1) \text{ unit} = 400 \text{ m}$$

$$\Rightarrow 1 \text{ unit} = \frac{400}{(\sqrt{3} + 1)} \times \frac{(\sqrt{3} - 1)}{(\sqrt{3} - 1)}$$

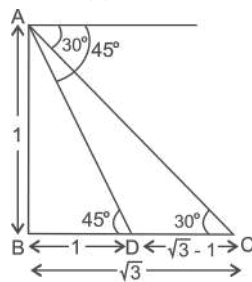
$$= 200(\sqrt{3} - 1) \text{ m}$$

Sol.30.(a)

$$\text{Width of river} = BD + DA = BA \Rightarrow \sqrt{3} + 1$$

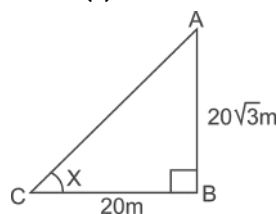
$$= 50(\sqrt{3} + 1) \Rightarrow 1 \text{ unit} = 50 \text{ m}$$

$$\text{So, the height of bridge} = 1 \text{ unit} = 50 \text{ m}$$

Sol.31.(c)

$$\text{According to question,} \\ 1 \text{ units} \rightarrow 60 \text{ m}$$

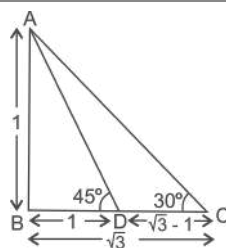
$$(\sqrt{3} - 1) \text{ units} \rightarrow 60 \times (\sqrt{3} - 1) \text{ m} \\ = 60 \times 0.732 = 43.92 \text{ m}$$

Sol.32.(c)

$$\text{In } \triangle ABC,$$

$$\Rightarrow \tan X = \frac{AB}{BC} = \frac{20\sqrt{3}}{20}$$

$$\Rightarrow \tan X = \sqrt{3} \Rightarrow X = 60^\circ$$

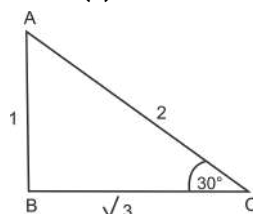
Sol.33.(b)

$$\text{According to question,}$$

$$(\sqrt{3} - 1) \text{ units} = 40 \text{ mts}$$

$$1 \text{ units} = \frac{40}{\sqrt{3} - 1} = 20(\sqrt{3} + 1) \text{ m}$$

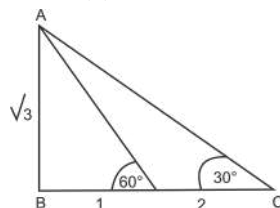
$$\text{Height of the tower} = 20(\sqrt{3} + 1) \text{ m}$$

Sol.34.(b)

$$A/Q, \sqrt{3} \text{ unit} = 70 \text{ m}$$

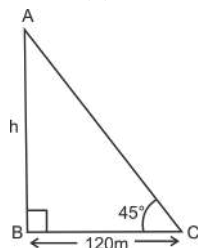
$$1 \text{ unit} = \frac{70}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{70\sqrt{3}}{3} \text{ m}$$

$$\therefore \text{Height of tower} = \frac{70\sqrt{3}}{3} \text{ m}$$

Sol.35.(d)

$$A/Q, 2 \text{ unit} = 200 \text{ m}$$

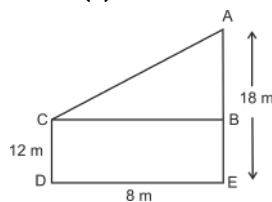
$$\sqrt{3} \text{ unit} = 100\sqrt{3} \text{ m}$$

Sol.36.(b)

$$\text{In } \triangle ABC,$$

$$\tan \theta = \frac{AB}{BC}$$

$$\tan 45^\circ = \frac{h}{120} \Rightarrow h = 120 \text{ mts}$$

Sol.37.(b)

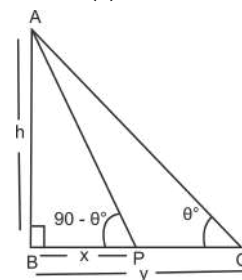
$$\text{Height difference (AB)}$$

$$= 18 \text{ m} - 12 \text{ m} = 6 \text{ m}$$

$$\text{Horizontal distance between poles (CB)} \\ = DE = 8 \text{ m}$$

$$\text{Distance between tops (AC)}$$

$$= \sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = 10 \text{ m}$$

Sol.38.(a)

$$\text{In } \triangle ABQ,$$

$$\tan \theta = \frac{h}{y} \text{ ---- (i)}$$

$$\text{In } \triangle ABP,$$

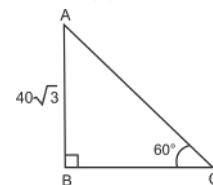
$$\tan(90^\circ - \theta) = \frac{h}{x}$$

$$\cot \theta = \frac{h}{x}$$

$$\tan \theta = \frac{x}{h} \text{ ---- (ii)}$$

$$\text{Equating the values of } \tan \theta:$$

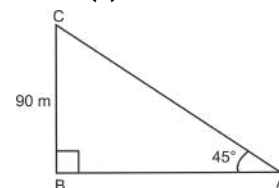
$$\frac{h}{y} = \frac{h}{x} \Rightarrow h = \sqrt{xy}$$

Sol.39.(a)

$$\text{In } \triangle ABC,$$

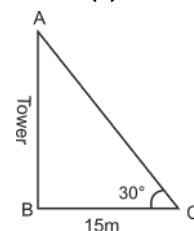
$$\Rightarrow \tan 60^\circ = \frac{40\sqrt{3}}{BC}$$

$$\Rightarrow \sqrt{3} = \frac{40\sqrt{3}}{BC} \Rightarrow BC = 40 \text{ m}$$

Sol.40.(d)

$$\text{In } \triangle ABC, \Rightarrow \sin 45^\circ = \frac{90}{AC}$$

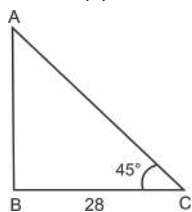
$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{90}{AC} \Rightarrow AC = 90\sqrt{2} \text{ m}$$

Sol.41.(d)

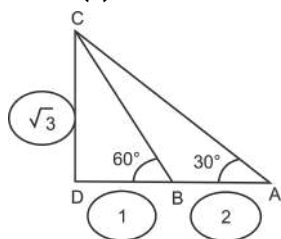
$$\text{In } \triangle ABC,$$

$$\Rightarrow \tan 30^\circ = \frac{AB}{15} \Rightarrow \frac{1}{\sqrt{3}} = \frac{AB}{15}$$

$$\Rightarrow AB = \frac{15}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 5\sqrt{3} \text{ m.}$$

Sol.42.(a)

$$\tan 45^\circ = \frac{AB}{28} \Rightarrow AB = 28 \text{ m}$$

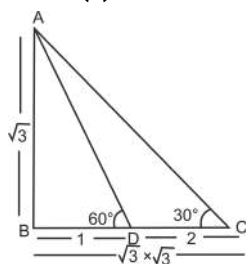
Sol.43.(d)

$$\text{Height } (\sqrt{3}) = 3 \text{ km} \Rightarrow 1 \text{ unit} = \sqrt{3} \text{ km}$$

$$2 \text{ unit} = 2\sqrt{3} \text{ km}$$

Speed of the airplane in km/hr

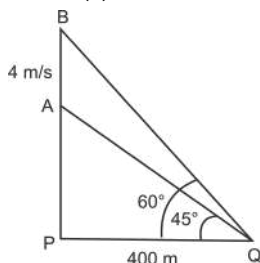
$$= \frac{2\sqrt{3} \text{ km}}{20 \text{ hr}} \times 3600 = 360\sqrt{3} \text{ km/hr}$$

Sol.44.(b)

$$\sqrt{3} \text{ units} = 1500\sqrt{3} \text{ meters}$$

$$2 \text{ units} = \frac{1500\sqrt{3}}{\sqrt{3}} \times 2 = 3,000 \text{ meters}$$

$$\text{Speed} = \frac{3,000}{15} = 200 \text{ m/s}$$

Sol.45.(a)

In $\triangle APQ$,

$$\Rightarrow \tan 45^\circ = \frac{AP}{400} \Rightarrow 1 = \frac{AP}{400}$$

$$\Rightarrow AP = 400 \text{ m}$$

In $\triangle BPQ$,

$$\Rightarrow \tan 60^\circ = \frac{PB}{400} \Rightarrow \sqrt{3} = \frac{AB + AP}{400}$$

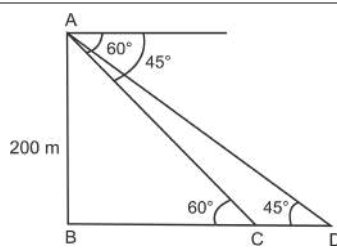
$$\Rightarrow 400\sqrt{3} = 400 + AB$$

$$\Rightarrow AB = 400\sqrt{3} - 400 = 400(\sqrt{3} - 1)$$

$$= 400(1.732 - 1) = 292.8 \text{ m}$$

So,

$$\text{Required time} = \frac{292.8}{4} = 73.2 \text{ seconds}$$

Sol.46.(a)

In $\triangle ABD$,

$$\Rightarrow \tan 45^\circ = \frac{200}{BD}$$

$$\Rightarrow 1 = \frac{200}{BD} \Rightarrow BD = 200 \text{ m}$$

In $\triangle ABC$,

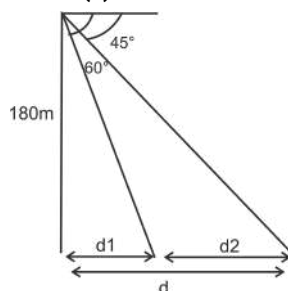
$$\Rightarrow \tan 60^\circ = \frac{200}{BC}$$

$$\Rightarrow \sqrt{3} = \frac{200}{BC} \Rightarrow BC = \frac{200}{\sqrt{3}}$$

Now,

Distance between two boats (CD) = BD - BC

$$\Rightarrow CD = 200 - \frac{200}{\sqrt{3}} = 200\left(\frac{\sqrt{3} - 1}{\sqrt{3}}\right) \text{ m}$$

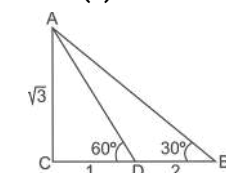
Sol.47.(c)

$$\tan 60^\circ = \frac{180}{d_1} \Rightarrow d_1 = \frac{180}{\sqrt{3}} = 103.92$$

$$\tan 45^\circ = \frac{180}{d_2} \Rightarrow d_2 = 180 \text{ m}$$

$$d_2 - d_1 = 180 - 103.92 = 76.08$$

$$\text{Speed} = \frac{76.08}{15} = 5.072 \text{ m/s}$$

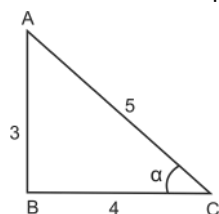
Sol.48.(b)

$$\text{Now, } A/Q \sqrt{3} \text{ unit} = 60\sqrt{3}$$

$$2 \text{ unit} = \frac{60\sqrt{3} \times 2}{\sqrt{3}} = 120 \text{ m}$$

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{120}{15} = 8 \text{ m/s}$$

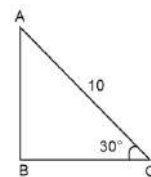
Sol.49.(c) Let AB and AC be the height of wall and ladder respectively



Here, 4 unit = 5

$$\text{Then, } 5 \text{ unit} = \frac{5}{4} \times 5 = \frac{25}{4} = 6.25 \text{ m}$$

Sol.50.(d) Let AB and AC be the length of wall and pole respectively.



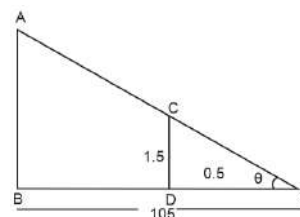
$$\text{In } \triangle ABC, \cos 30^\circ = \frac{BC}{10}$$

$$\frac{\sqrt{3}}{2} = \frac{BC}{10}$$

$$BC = \frac{10\sqrt{3}}{2} = 5\sqrt{3} \text{ m}$$

So, the foot of the pole is $5\sqrt{3}$ m away from the wall.

Sol.51.(c) Let AB be the height of tower and CD be the height of person and Mirror is placed at O.

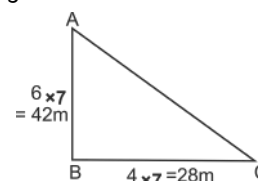


$\triangle OCD \sim \triangle OAB$ (By AAA rule)

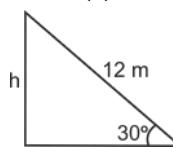
$$\frac{CD}{AB} = \frac{OD}{OB} \Rightarrow \frac{1.5}{AB} = \frac{0.5}{105}$$

$$AB = \frac{1.5 \times 105}{0.5} = 315 \text{ m}$$

Sol.52.(d) Let AB be the length of the pole and BC be the shadow on the ground.

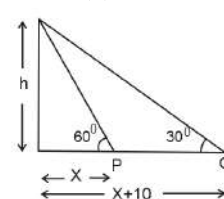


So, the height of tower = 42 m

Sol.53.(d)

$$\sin 30^\circ = \frac{h}{12}$$

$$\frac{1}{2} = \frac{h}{12}, h = 6 \text{ m}$$

Sol.54.(a)

$$\tan 60^\circ = \frac{h}{x}$$

$$h = \sqrt{3}x \dots\dots\dots(i)$$

$$\tan 30^\circ = \frac{h}{x+10}$$

$$\sqrt{3}h = x + 10$$

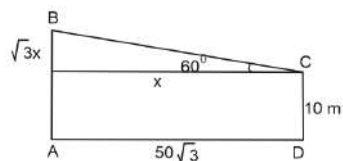
$$h = \frac{x+10}{\sqrt{3}} \dots\dots(ii)$$

From equations (i) and (ii) we get

$$\sqrt{3}x = \frac{x+10}{\sqrt{3}},$$

$$\therefore x = 5 \Rightarrow h = 5\sqrt{3} \text{ m}$$

Sol.55.(c) Let AB and CD be the height of tower and building respectively.



$$\text{Height of tower} = (\sqrt{3}x + 10)\text{m}$$

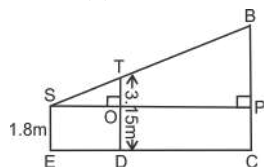
$$\text{ATQ, } x = 50\sqrt{3}$$

$$\text{Then, height of tower} = (\sqrt{3} \times 50\sqrt{3} + 10) \\ = 150 + 10 = 160\text{m}$$

Sol.56.(d)

$$\frac{42}{35} = \frac{x}{25} \Rightarrow x = \frac{42 \times 25}{35} = 30 \text{ m}$$

Sol.57.(a)



$$TO = TD - OD = 3.15 - 1.8 = 1.35 \text{ m}$$

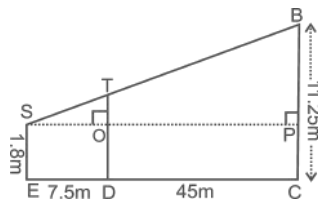
$$\triangle SOT \cong \triangle SPB$$

$$\frac{SO}{SP} = \frac{TO}{BP} \Rightarrow \frac{7.5}{45 + 7.5} = \frac{1.35}{BP}$$

$$BP = \frac{1.35 \times 52.5}{7.5} = 9.45 \text{ m}$$

$$\text{Height of the building} = (BP + PC) \\ = 9.45 + 1.8 = 11.25 \text{ m}$$

Sol.58.(c)



$$PB = BC - PC = 11.25 - 1.8 = 9.45 \text{ m}$$

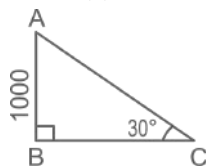
$$\triangle SOT \cong \triangle SPB$$

$$\frac{SO}{SP} = \frac{TO}{BP} \Rightarrow \frac{7.5}{7.5 + 45} = \frac{TO}{9.45}$$

$$TO = \frac{7.5 \times 9.45}{52.5} = 1.35 \text{ m}$$

$$\text{Length of tree (TD)} = TO + OD \\ = 1.35 + 1.8 = 3.15 \text{ m}$$

Sol.59.(c)



$$\tan 30^\circ = \frac{1000}{BC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{1000}{BC}$$

$$BC = 1000\sqrt{3} \text{ m} = 1000 \times 1.732 = 1732 \text{ m}$$

Alternate method :

Ratio of length of side corresponding to

$$30^\circ, 60^\circ \text{ and } 90^\circ = 1 : \sqrt{3} : 2$$

$$1 \text{ unit} = 1000 \text{ m}$$

(distance between building and man)

$$\sqrt{3} \text{ units} = 1000 \times 1.732 = 1732 \text{ m}$$

Sol.60.(a)

Shadow of pole $\propto$ height of pole

A pole of 28 m height casts a shadow of 19.2 m long .

52.5 m long pole casts a shadow of

$$= \frac{19.2}{28} \times 52.5 = 36 \text{ metre}$$